INSY 8311 · DATABASE DEVELOPMENT WITH PL/SQL · PHASE 0

VoltGuard Rwanda

Electricity Billing and Meter Tamper Monitor

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WHY IT MATTERS

Background and problem

Tariffs change

RURA revised the end-user tariff on 1 October 2025. The first residential band grew from 15 to 20 kWh.

Readings can be wrong

Faulty or manipulated readings and billing errors are non-technical losses (Glauner et al., 2016).

Rules are hard to see

Public information explains how to buy prepaid electricity, not how readings are validated and audited.

WHO IS AFFECTED

The utility loses revenue, and customers receive wrong bills.

Sources: RURA (2025); Glauner et al. (2016); Rwanda Energy Group (n.d.)

MOTIVATION

A wrong bill costs the utility revenue and costs the customer trust

I want to see how the rules behind a bill can be enforced inside the database, so that a wrong reading or an outdated tariff cannot go unnoticed.

WHAT WE WILL BUILD

Objectives

MAIN OBJECTIVE

Build an Oracle PL/SQL database that bills readings with the correct versioned tariff and rejects and logs suspicious readings.

1. Normalise real consumption data and RURA tariffs into 3NF tables
2. Bill each reading with the tariff version valid on its date
3. Reject any reading lower than the previous one and log the attempt
4. Audit every change to readings, bills and tariffs
5. Give staff and customers an app to enter readings and look up bills

LITERATURE AND EXISTING SYSTEMS

What exists, and what is missing

Glauner et al. (2016)	Non-technical losses, such as faulty readings and billing errors, are a known problem for power utilities.
RURA (2025)	The official tariff notice, effective 1 October 2025, sets the bands and prices I will load as data.
REG FAQ (n.d.)	The existing system: customers can learn how to buy prepaid electricity.

THE GAP

Nothing public shows how readings are validated, how old tariffs are kept, or how changes are audited.

DATA SOURCE

Real minute-by-minute readings

2,075,259
measurements, 2006 to 2010

1 minute
between readings

CC BY 4.0
free to reuse with credit

Date	Time	Global_active_power	Voltage	Global_intensity
16/12/2006	17:24:00	4.216	234.840	18.400
16/12/2006	17:25:00	5.360	233.630	23.000
16/12/2006	17:26:00	5.374	233.290	23.000
16/12/2006	17:27:00	5.388	233.740	23.000
16/12/2006	17:28:00	3.666	235.680	15.800

First rows of the UCI file (Hebrail and Berard, 2012). Power is in kilowatts, so kWh per minute = Global_active_power / 60.

Real: the readings and the RURA tariffs

Simulated: customers, meters and districts

PROPOSED SOLUTION

VoltGuard keeps the rules in the database

- **Validates:** a trigger rejects a reading lower than the previous one
- **Bills:** a function applies the tariff version valid on the reading date
- **Audits:** every attempt and change is logged, even a rejected one

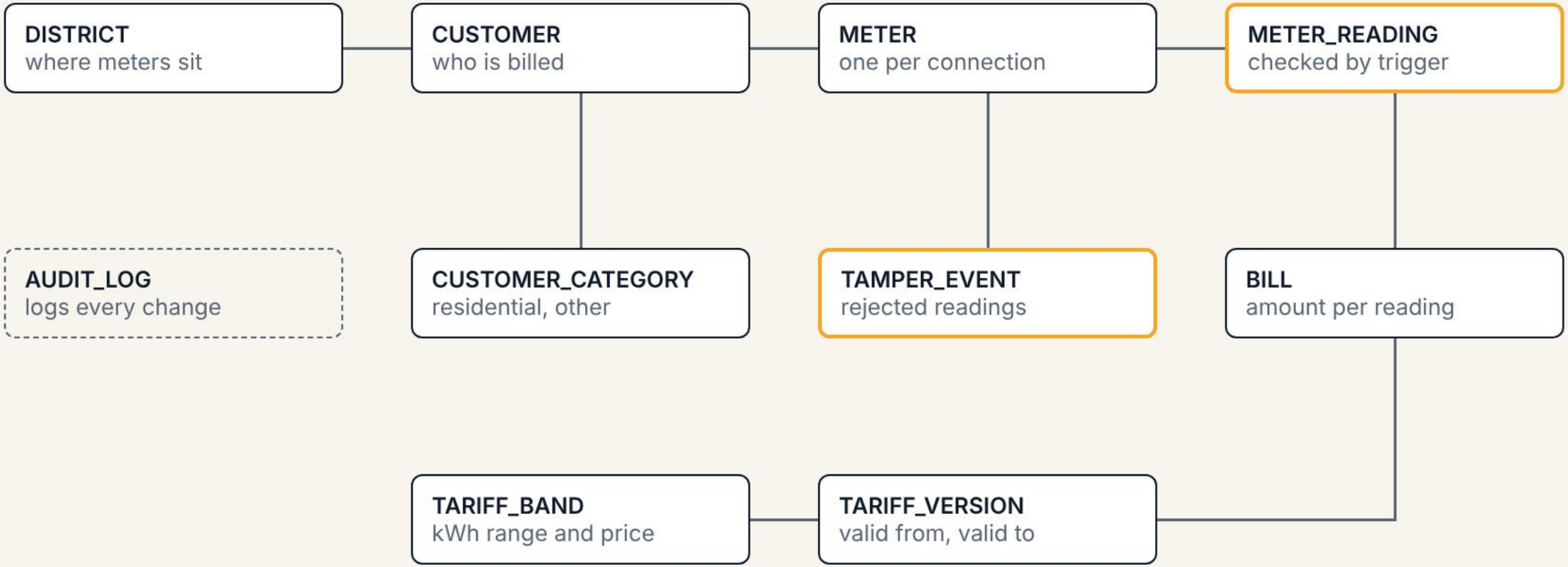
INNOVATION: USER INTERFACE

Oracle APEX app

Staff enter readings and see rejections. A REST bill lookup (ORDS) serves customers.

PLANNED TABLES

Database design in 3NF



Each line is a one-to-many link. Amber boxes are guarded by triggers.

PLANNED PL/SQL FEATURES

Where each feature is used

Feature	Where used
Trigger	Reject a reading lower than the previous one
Autonomous transaction	Keep the audit log even when a reading is rolled back
Function	Find the tariff version valid on a date and compute a bill
Procedure	Post a reading and create its bill
Cursor	Loop over a meter’s readings to build a statement
Custom exception	Signal an invalid reading with a clear error
Package	Group the billing code in one place

LIVE DEMO PLAN

What you will see

1. Enter a normal reading and see it billed
2. Enter a lower reading and see it rejected and logged
3. Show that an older date uses the older tariff version
4. Look up the bill through the REST link

EXAMPLE BILL, 2025 RESIDENTIAL

30 kWh = 20 × 89 + 10 × 310

4,880 RWF

Before VAT and the regulatory fee

HONEST SCOPE

Limitations and next steps

Limitations

The readings come from one French household, so customers, meters and districts are simulated and consumption is scaled.

Prices exclude VAT and the regulatory fee.

Only the residential category is billed in the demo.

Future improvements

Load real smart-meter readings instead of simulated ones.

Add VAT and the regulatory fee to every bill.

Add more tamper rules, such as a sudden unexplained drop in use.

SOURCES

References

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Full links are in docs/research/research_log.md on GitHub.